Alexander DeLise

Email: alex.r.delise@gmail.com
Google Scholar: Alexander DeLise
Personal Website: alexdelise.com
LinkedIn: alexanderdelise
GitHub: github.com/alexdelise

Education

Florida State University — Tallahassee, FL
B.S. Computational Science with Honors, GPA: 4.00/4.00 — 2023–2027

- Thesis: Active Learning for Conditional Generative Compressed Sensing
- Advised by Dr. Nick Dexter through the Honors in the Major Program

Florida State University — Tallahassee, FL
B.S. Applied and Computational Mathematics, GPA: 4.00/4.00 — 2023–2027

- Activities: Presidential Scholars Program, Undergraduate Research Opportunity Program, *President*, Pi Mu Epsilon Mathematics Honor Society, Society for Industrial and Applied Mathematics

Research Experience

Florida State University — Tallahassee, FL
Honors Thesis — Advised by Dr. Nick Dexter — November 2025 – Present

- Research topics: compressed sensing, generative modeling, inverse problems, active learning, scientific machine learning, representation learning, imaging
- Developed sample complexity theory for conditional generative models under ReLU / Lipschitz assumptions for stable recovery
- Introduced a compatibility factor that quantifies sample complexity and signal recovery penalties for mismatches in conditioning
- Performed extensive image recovery experiments with Stable Diffusion across 15+ NVIDIA A5000 GPUs, showcasing how conditioning affects sampling distributions and recovery
- Languages and tools: Python, PyTorch, Hugging Face, Slurm, Tmux, Bash

Emory University — Atlanta, GA
Computational Math for Data Science REU — Advised by Dr. Matthias Chung — June 2025 – August 2025

- Research topics: scientific machine learning, inverse problems, low-rank matrix approximation, Bayes risk minimization
- Developed a unified Bayes-risk framework and derived closed-form, rank-constrained linear encoder-decoder mappings for forward modeling and inverse recovery
- Generalized optimal mappings to rank-deficient data, operators, and measurements for realistic scientific machine learning settings
- Demonstrated interpretable, competitive baselines in nonlinear problem settings on imaging, finance, and PDE datasets
- Languages and tools: Python, PyTorch

University of Tennessee, Knoxville — Knoxville, TN
Quantum Algorithms and Optimization REU — Advised by Dr. James Ostrowski — May 2024 – April 2025

- Research topics: quantum computing, variational quantum algorithms, quantum optimization
- Developed Partitioned-Constraint QAOA, a hybrid framework that structurally enforces selected constraints through feasible-state preparation and Grover mixing while penalizing the remaining constraints

- Trained reusable variational constraint gadgets for arbitrary low-support constraints and demonstrated improved shallow-depth feasibility and solution quality over penalty-based QAOA
- Languages and tools: Python, PennyLane, PyTorch, Slurm

FAMU-FSU College of Engineering Tallahassee, FL
Operations Research — Advised by Dr. Arda Vanli September 2023 – June 2025

- Research topics: operations research, industrial engineering, mixed-integer nonlinear programming, data-driven healthcare, disease modeling
- Developed a data-driven mixed-integer nonlinear programming model that couples SIRV epidemic dynamics with patient movement
- Tested Florida COVID-19 data to demonstrate how vaccination and mobility jointly affect surge patterns and unmet hospital bed demand
- Languages and tools: Python, Gurobi, GAMS, IBM ILOG CPLEX

Industry Experience

Johns Hopkins University Applied Physics Laboratory Laurel, MD
Data Science Intern May 2026 – August 2026

- **Machine Learning:** Built end-to-end pipeline for training neural operator ML models to predict aerodynamic properties of hypersonic nose-cone geometries, spanning data generation to Slurm-managed model training on CPU cluster
- Led ablation study using a statistical design-of-experiments approach and paired noninferiority testing to evaluate the tradeoff between model size and predictive accuracy across reduced model configurations
- Benchmarked 25+ model variants, identifying reduced configurations that preserved statistically noninferior accuracy while reducing model parameter counts by up to 85% and training time by over 14×
- **Optimization & Decision Science:** Developed a multi-site equipment transfer optimization framework in MATLAB, integrated within a system dynamics logistics model, with an efficient ∼10 second runtime
- Evaluated framework on real data, reducing equipment shortfall by up to 50% and transfers by up to 61%
- Calculated Pareto front of equipment shortfall vs. transfer volume to support readiness and logistics decisions

Publications

Journal Articles

1. **A. DeLise**, K. Loh, K. Patel, M. Teague, A. Arnold, M. Chung. "Optimal Linear Baseline Models for Scientific Machine Learning." *Foundations of Data Science*. [doi] [arXiv]

Conference Proceedings

1. **A. DeLise**, S. Abazari, A. Vanli. "An Epidemiological Mixed-Integer Nonlinear Programming Framework for Vaccine Modeling and Patient Allocation During Pandemics." *International Conference on Industrial Engineering and Operations Management*. [doi]

Preprints

1. **A. DeLise**, N. Dexter. "Active Learning for Conditional Generative Compressed Sensing." 2026. [arXiv]
2. A. Wilkie, **A. DeLise**, A. Del Real, R. Herrman, J. Ostrowski. "Partitioned-Constraint QAOA (PC-QAOA): Structural State Preparation and Penalty Enforcement for Quantum Optimization." 2026. [arXiv]

Presentations

Contributed Talks

1. Active Learning for Conditional Generative Compressed Sensing [pdf]
   April 2026. ACC Meeting of the Minds, Tallahassee, FL.
2. Optimal Linear Baseline Models for Scientific Machine Learning [pdf]
   January 2026. Joint Mathematics Meetings, AMS Contributed Paper Session on Computer Science, Information, and Communication, Washington, DC.
3. Optimal Linear Baseline Models for Scientific Machine Learning [pdf]
   November 2025. SIAM Student Chapter, Florida State University, Tallahassee, FL.

Poster Presentations

1. Active Learning for Conditional Generative Compressed Sensing [pdf]
   April 2026. 26th Annual Undergraduate Research Symposium, Florida State University, Tallahassee, FL.
2. Optimal Rank-Constrained Mappings for Linear ED Architectures [pdf]
   March 2026. Florida Undergraduate Research Conference, University of North Florida, Jacksonville, FL.
3. Optimal Rank-Constrained Mappings for Linear ED Architectures [pdf]
   January 2026. Joint Mathematics Meetings AMS-PME Student Poster Session, Washington, DC.
4. Optimal Rank-Constrained Mappings for Linear ED Architectures [pdf]
   August 2025. CMDS REU Poster Presentation, Emory University, Atlanta, GA.
5. Data-Driven Patient Allocation Optimization with Epidemic and Vaccine Modeling [pdf]
   April 2025. 25th Annual Undergraduate Research Symposium, Florida State University, Tallahassee, FL.
6. Data-Driven Patient Allocation Optimization with Epidemic and Vaccine Modeling [pdf]
   February 2025. Florida Undergraduate Research Conference, University of South Florida, Tampa, FL.
7. Learning Feasible States with ma-QAOA and QAOA for Constrained Optimization [pdf]
   January 2025. Joint Mathematics Meetings AMS-PME Student Poster Session, Seattle, WA.
8. Learning Feasible States with ma-QAOA and QAOA for Constrained Optimization [pdf]
   July 2024. UTK Summer Research Symposium, University of Tennessee, Knoxville, TN.
9. Cost-Effective Location Allocation for COVID-19 Patient Assignment in Florida: A Data-Driven Approach [pdf]
   April 2024. 24th Annual Undergraduate Research Symposium, Florida State University, Tallahassee, FL.

Press and Media Coverage

1. Florida State University News. "FSU achieves historic milestone: Four students named 2026 Goldwater Scholars." April 2026. [link]

Honors and Awards

- Barry M. Goldwater Scholar — 2026
- Pi Mu Epsilon Travel Grant — 2025, 2026
- FURC Travel Grant, Florida State University — 2025
- Presidential Scholar, Florida State University — 2023
- Bright Futures Academic Scholar, State of Florida — 2023